



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SEMESTER 1

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SECJ3553-02 ARTIFICIAL INTELLIGENCE (AI)

ASSIGNMENT 3

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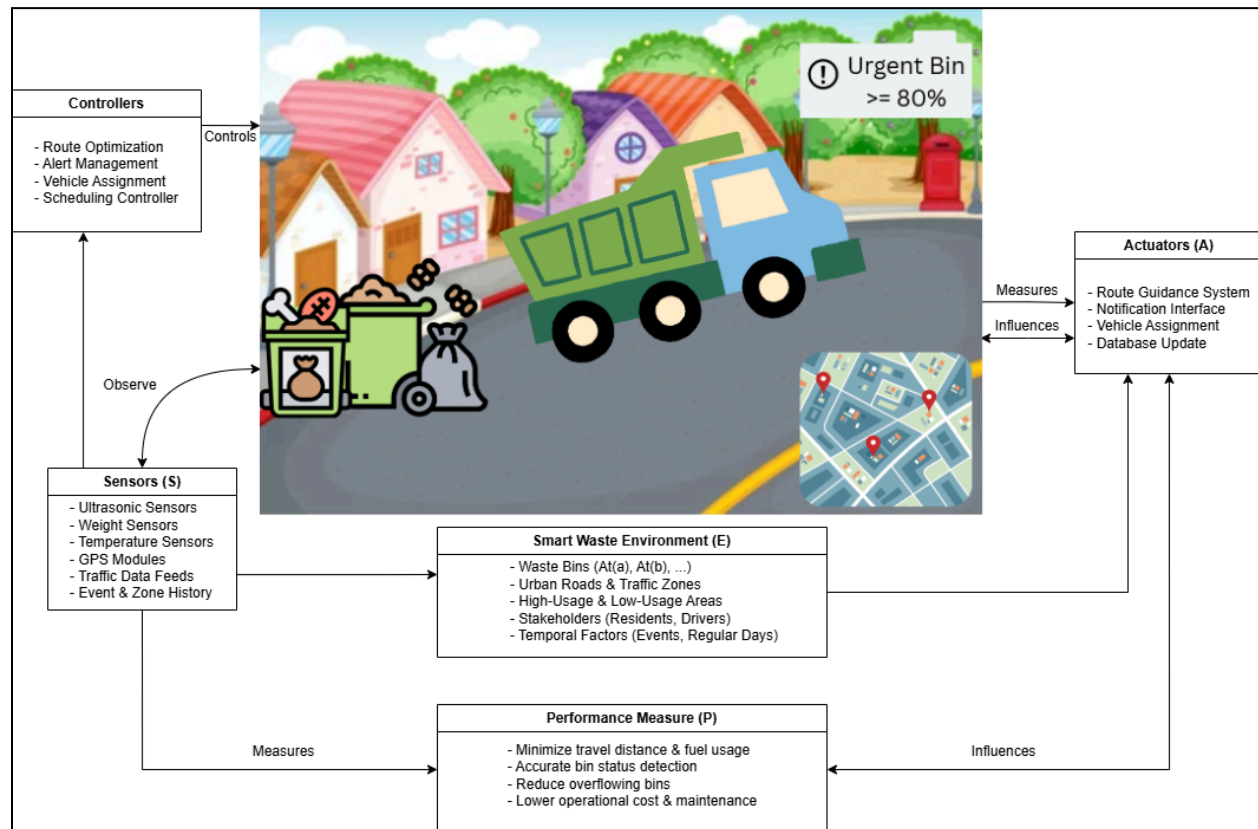
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1.0 PEAS MODEL

Agent	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Smart Waste Management Agent	Efficiency & Accuracy: Minimize total distance traveled and fuel consumption. Correct identification of Medium , Near-Full ($\geq 70\%$) , and Critical ($\geq 80\%$) bins .	Urban Layout & Zones: Roads, intersections, and traffic zones. Divided into High-Usage Zones (fast accumulation) and Low-Usage Zones .	Route Guidance System: Displaying optimized paths (shortest route) to the driver.	Hardware Sensors; Ultrasonic Sensors: To detect real-time fill height. Weight Sensors: To measure the load/mass of the waste. Temperature Sensors: To detect potential fire hazards.
	Cleanliness & Hygiene: Minimize the number of overflowing bins (collection latency), unpleasant odors, and pests.	Waste Bins: Located at various static coordinates (e.g., At(a), At(b)) with changing fill states.	Notification & Alert Interface: Issuing “Urgent Collection Requests” to the central dashboard or driver when bins reach the $\geq 70\%$ threshold.	Location & Tracking; GPS Modules: To track the real-time location of collection trucks.
	Cost & Sustainability: Reduce operational costs (labor hours, vehicle wear) and carbon emissions.	Stakeholders: Residents (waste generators) and Waste Collection Workers (drivers).	Vehicle Assignment: Electronically assigning the nearest available vehicle to an urgent bin.	Data Inputs; Traffic Data Feeds: Real-time density information. Event Calendar: To detect special event days.
	Response Time: Ensure urgent bins are detected and serviced	Temporal Factors: Event Days (holidays/festivals)	Database Updates: Logging collected bins, route completion	Zone Usage

	promptly by the nearest vehicle.	vs. Regular Days affecting waste generation rates	status, and “Safe” status after collection.	History: To predict accumulation trends.
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2.0 Relations Between Each Property



3.0 Explanation In Proof Of Concept (POC)

• Performance

The agent's performance is measured by its ability to optimize waste collection logistics and maintain urban hygiene. It aims to minimize total distance travelled and fuel consumption by providing optimized route guidance to drivers. Success is also defined by the system's accuracy in identifying bin fill levels which are categorized as Medium, Near-Full ($\geq 70\%$), and Critical ($\geq 80\%$) to prevent overflows, odors, and pest issues.

• Environment

The environment comprises the urban infrastructure and the stakeholders involved in waste generation and collection. This includes a network of roads, intersections, and traffic zones divided into high-usage and low-usage areas. The waste bins are located at static coordinates, such as $At(a)$ and $At(b)$, while temporal factors like holidays and festivals influence the rate of waste accumulation.

- **Actuators**

Actuators allow the agent to affect the environment and manage logistics. The Route Guidance System displays optimized paths to drivers, while the Notification & Alert Interface issues urgent collection requests when bins hit the 70% threshold. Additionally, the system performs Vehicle Assignment by electronically dispatching the nearest truck and updates the Database to log completed collections and “Safe” area statuses.

- **Sensors**

The agent perceives its surroundings through various hardwares and data inputs. Ultrasonic sensors detect real-time fill height, weight sensors measure the mass of waste, and temperature sensors monitor for fire hazards. For tracking, GPS modules provide truck locations, while Traffic Data Feeds and Event Calendars help the agent anticipate road density and seasonal waste spikes.

4.0 Behaviors Of Agent To Achieve The System Goals

The Smart Waste Management Agent is a goal-based agent. Its primary goal is to maximize collection efficiency while ensuring no bins reach an overflowing state.

The agent interacts with the environment by continuously monitoring sensor data. Once active, the sensors detect the fill level of bins across various zones. The agent processes this information to compute the most efficient collection path, prioritizing bins at the critical $\geq 80\%$ threshold. For example, if a bin in a high-usage zone becomes full, the agent immediately alerts the driver and adjusts their route.

The agent also updates the central dashboard with real-time zone status and historical usage trends. When a bin reaches the 70% “urgent” threshold, the agent automatically assigns the nearest available vehicle to service that location. Throughout the process, the agent assists the driver by providing navigation that avoids heavy traffic. After collection, the agent marks the bin as “Safe”, ensuring consistent improvement in urban cleanliness and operational sustainability.